Big,little exercise

"Ex Post" Global changes: $f_2 \to f$, $h_2 \to h$ } simplifies notation

Planning problem:

Criterion: $c_t + u_t$ ~ period criterion

upsilon

means not there

cross out

$$c_t + u_t = \cancel{f_0} + s_t n_t - \frac{f_2}{2} n_t^2 - \frac{d}{2}(n_{t+1} - n_t)^2$$
$$+ \cancel{h_0} - b_t n_t - \frac{h_2}{2} n_t^2 - \frac{g}{2}(n_{t+1} + n_t)^2$$

Experiment with state:

set to zero - put constants into s_t, b_t

$$u_t = n_{t+1} - n_t \Rightarrow$$

$n_{t+1} = n_t + u_t$, $n_{t+1} + n_t = n_t + u_t + n_t = 2n_t + u_t$

$$n_{t+1} + n_t \overset{?}{=} n_t + u_t + n_t = 2n_t + u_t =$$
$$= n_t + (n_{t+1} - \cancel{n_t}) + \cancel{n_t} \Rightarrow$$

$$n_{t+1} + n_t = 2n_t + u_t$$

$$\therefore (n_{t+1} + n_t)^2 = (2n_t + u_t)^2 = 4n_t^2 + 4n_t u_t + u_t^2$$

$$\therefore c_t + u_t = (\cancel{f_0} + \cancel{h_0}) + (s_t - b_t) n_t - \frac{(f_2 + h_2)}{2} n_t^2$$
$$- \frac{d}{2} u_t^2 - \frac{g}{2}(4n_t^2 + 4n_t u_t + u_t^2)$$

upsilon

$$c_t + u_t = \cancel{(f_0 + h_0)} + (s_t - b_t)\, n_t$$

$$- \left(\frac{f_1 + h_2 + 4q}{2}\right) n_t^2 - \left(\frac{d+q}{2}\right) u_t^2$$

$$- 4q\, n_t u_t$$

$$\left.\begin{aligned} s_t &= S_s z_t \\ b_t &= S_b z_t \end{aligned}\right\} \text{ is } \quad \underbrace{\cancel{f_0 = S_s z_t}, \ \cancel{h_0 = S_h z_t}}_{=0}$$

$$c_t + u_t = \cancel{(f_0 + h_0)} + \overbrace{(S_s - S_b)}^{s_t - b_t} z_t\, n_t$$

$$- \left(\frac{f_2 + h_2 + 4q}{2}\right) n_t^2 - \left(\frac{d+q}{2}\right) u_t^2$$

$$- 4q\, n_t u_t$$

$$c_t + u_t = \cancel{(S_s + S_h)} z_t + (S_s - S_b) z_t\, n_t$$

$$- \left(\frac{f_2 + h_2 + 4q}{2}\right) n_t^2 - \left(\frac{d+q}{2}\right) u_t^2$$

$$- 4q\, n_t u_t$$

$$= \begin{pmatrix} z_t \\ n_t \end{pmatrix}' \overbrace{\begin{pmatrix} 0 & \left(\frac{S_s - S_b}{2}\right)' \\ \left(\frac{S_b - S_s}{2}\right), & -\left(\frac{f_2 + h_2 + 4q}{2}\right) \end{pmatrix}}^{R} \begin{pmatrix} z_t \\ n_t \end{pmatrix} - \overbrace{\left(\frac{d+q}{2}\right)}^{Q} u_t^2$$

$$- 4u_t \underbrace{\begin{bmatrix} 0 & q \end{bmatrix}}_{W} \begin{bmatrix} z_t \\ n_t \end{bmatrix}$$

R, Q, W not signs in front of above

Tom's

~~Be careful about the P's in vector.~~ ~~See below~~

~~Check chapter 5 carefully first.~~

$$\begin{pmatrix} z_{t+1} \\ n_{t+1} \end{pmatrix} = \begin{pmatrix} A_{22} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z_t \\ n_t \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u_t + \begin{pmatrix} C_2 \\ 0 \end{pmatrix} \varepsilon_{t+1}$$

firm problem:

set $f_0 = 0$:

$$\Pi_t = S_t n_t - \frac{f_2}{2} n_t^2 - \frac{d}{2}(n_{t+1} - n_t)^2 - w_t n_t$$

$$\begin{pmatrix} s_t \\ w_t \end{pmatrix} = \begin{pmatrix} S_s \\ S_w \end{pmatrix} z_t$$

$$\pi_t = (S_s - S_w)\, n_t - \frac{f_2}{\cancel{\pi_t}}\, n_t^2 - \frac{d}{\cancel{\pi_t}\,2}\, u_t^2$$

$$n_{t+1} = n_t + u_t$$

$$\begin{pmatrix} z_{t+1} \\ n_{t+1} \end{pmatrix} = \begin{pmatrix} A_{22} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z_t \\ n_t \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u_t + \begin{pmatrix} C_2 \\ 0 \end{pmatrix} \varepsilon_{t+1}$$

$$\pi_t = \begin{pmatrix} z_t \\ n_t \end{pmatrix}' \begin{pmatrix} 0 & \frac{(S_s - S_w)}{2}' \\ \frac{(S_s - S_w)}{2} & -\frac{f_2}{\cancel{\pi_t}} \end{pmatrix} \begin{pmatrix} z_t \\ n_t \end{pmatrix} - \frac{d}{\cancel{\pi_t}\,2}\, u_t^2$$

Reformulate with Big K - little k // Firm problem

$$\begin{pmatrix} z_{t+1} \\ \tilde{n}_{t+1} \\ \hat{n}_{t+1} \end{pmatrix} = \left(\begin{array}{c|c} \tilde{A} - \tilde{B}\tilde{F} & 0 \\ \hline 0 & 1 \end{array}\right) \begin{pmatrix} z_t \\ \tilde{n}_t \\ \hat{n}_t \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \hat{u}_t$$

$$+ \begin{pmatrix} \tilde{C} \\ \hline 0 \end{pmatrix} \varepsilon_{t+1}$$

$$s_t = \begin{bmatrix} S_s & 0 \end{bmatrix} \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix} \equiv \tilde{S}_s \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix}$$

$$b_t = \begin{bmatrix} S_b & 0 \end{bmatrix} \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix} \equiv \tilde{S}_b \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix}$$

$2P_n$

$$w_t = \begin{bmatrix} 2P_n \end{bmatrix} \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix} \equiv \tilde{S}_w \begin{bmatrix} z_t \\ \tilde{n}_t \end{bmatrix}$$

$$\pi_t = (S_s - S_w) \begin{pmatrix} z_t \\ \tilde{n}_t \end{pmatrix} \hat{n}_t - f_2 \hat{n}_t^2 - d \hat{u}_t^2$$

$$\pi_t = \begin{pmatrix} z_t \\ \tilde{n}_t \\ \hline \hat{n}_t \end{pmatrix} \underbrace{\left(\begin{array}{c|c} 0 & \frac{(S_s - S_w)'}{2} \\ \hline \frac{(S_s - S_w)}{2} & -f_2 \end{array}\right)}_{\hat{R}} \underbrace{\begin{pmatrix} z_t \\ \tilde{n}_t \\ \hline \hat{n}_t \end{pmatrix}}_{\hat{x}_t} - \underbrace{d}_{\hat{Q}} \hat{u}_t^2$$

$$\hat{\ } \quad \hat{E}\hat{\ } \quad \hat{\ } - \begin{pmatrix} z_t \end{pmatrix}$$

$u_t = -F x_t, \quad x_t = \begin{pmatrix} \tilde{n}_t \\ \hat{n}_t \end{pmatrix}$

Scratch work:

value functions ~ scalars

if $x_t' P x_t = W(x_t)$

then $\dfrac{\partial W(x_t)}{\partial x_t} = 2 P x_t$

$$-x_t P x_t = \max_{u_t} \{ -x_t' R x_t - u_t' Q u_t - \beta x_{t+1}' P x_{t+1} \}$$

$$x_{t+1} = A x_t + B u_t$$

$$\frac{\partial}{\partial x_t} V(x_t) = -2 P x_t$$

probably want a 2

Last modified: 1:52 AM